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Preface

About

This document contains detailed explanation about how to integrate with PayFast API's Based transactions functionality. This document also contains the details for online transactions.

Intended Audience

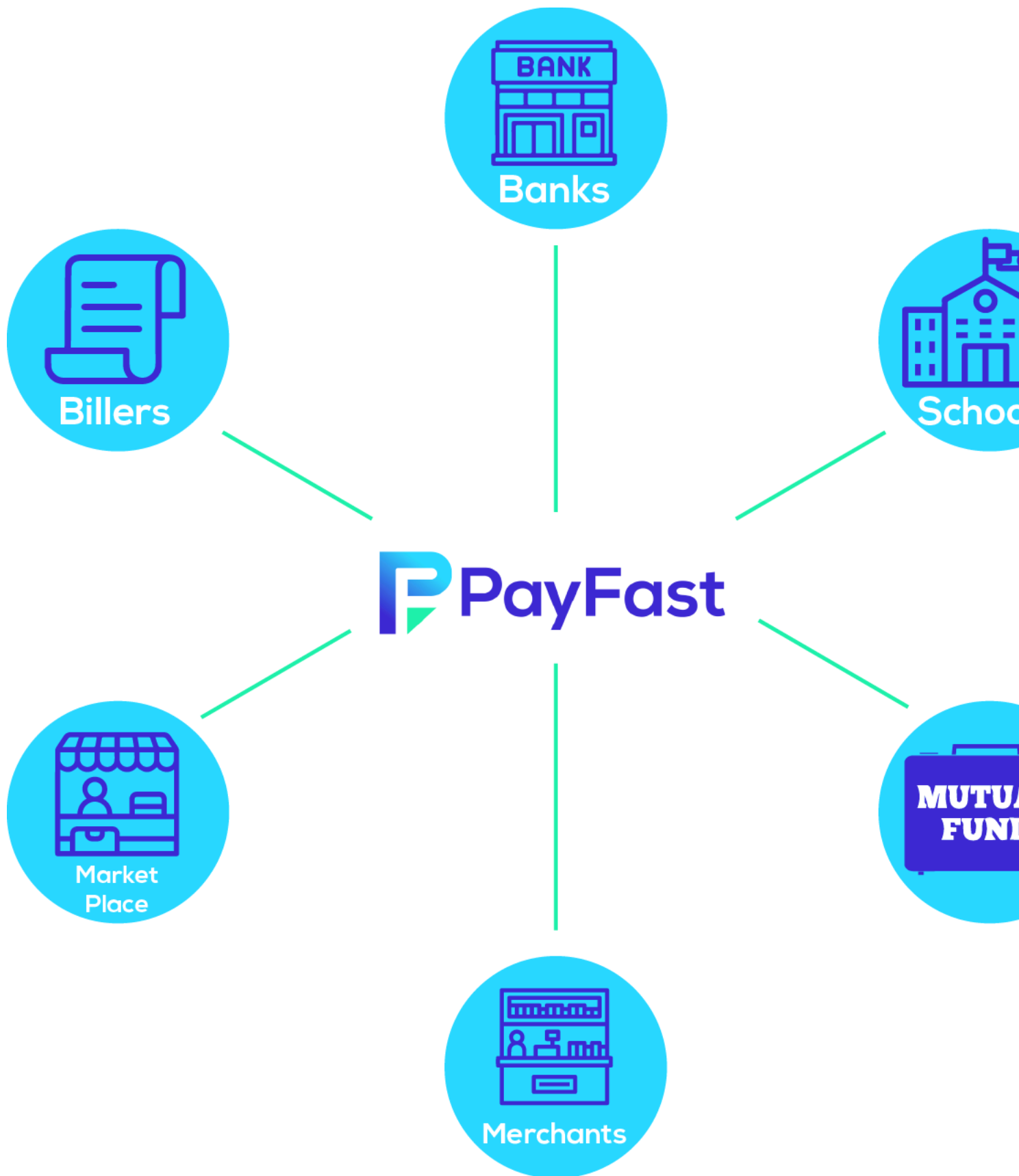
This document is for merchants, acquirers and developers who want to integrate with PayFast to perform a API's based Transactions.

Integration scope

The merchant will implement all ecommerce functionality. PayFast' service (Payfast) will be used only for payment processing.

PayFast – Go Cashless!

PayFast is focused to bring merchants on ecommerce platforms by offering a complete suite of solutions to all types of merchants from building a website, online store, mobile application, social media marketing to securing payment. PayFast will enable merchants to sell the products and services online, and receive their payment safely.



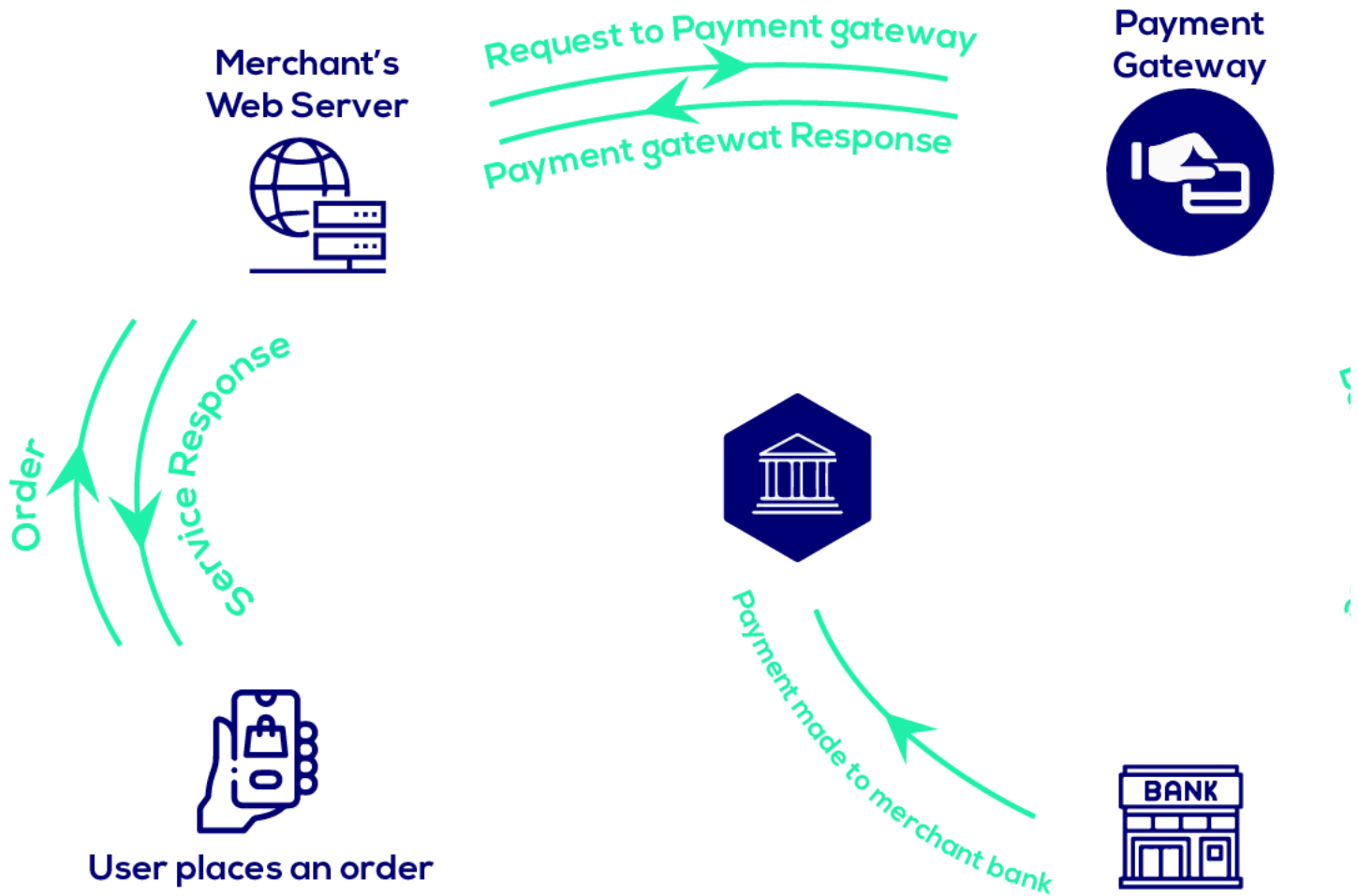
Key Features

- Simple, secure and low risk
- Offers easy integration and a broad application suite
- Directly charge from the customer's Bank account, Debit Card and Wallets
- Guaranteed prompt payment to Merchants

- Integrates easily with Mobile and web based application
- Connects to a variety of Merchant organizations including retail stores, governments, etc.

How Payfast Works

PayFast has developed a flagship payment gateway solution called "PayFast" which is targeted towards merchants, schools and other corporate entities to accept payments digitally from customers by using multiple options such as Debit Cards and Account Numbers. It will be integrated directly with the client's website to allow smooth transaction placements.



API Based Transaction

All retail stores can integrate with Payfast direct API's for over the counter payments without redirection. After the payment process is completed, the merchant will get an intimation of the transaction status.

Integration Prerequisites

Merchants will be registered on PayFast prior to integration. After merchant sign up for PayFast account, following two unique values will be provided to merchant to operate: * MERCHANT_ID * SECURED_KEY. These keys are used to get a one-time authentication token, which is used to authenticate payment requests to the "Payfast" payment gateway.

Getting Authentication Token

To initiate the transaction, the merchant needs to generate the Authentication Token from PayFast server. For fetching, the token developer needs to call the Authentication token API.

Let's start with Scenarios

SCENARIO 1

Consider a typical ecommerce transaction where a customer enters card/account details and tries to pay via PayFast. In this scenario, a temporary transaction token (refer to this section) should be fetched (which will last for short period) and then a tokenized transaction API (refer to this section) will be called to initiate a payment request based on temporary token. In this process, an OTP will be sent to customer mobile number. This token cannot be used for recurring transaction.

SCENARIO 2

Consider a scenario where a mobile app wants its customer card information for monthly subscription. This is recurring model, where customer will store its instrument such as card details or account details for later use. It is recommended for merchant to get a temporary transaction (refer to this section) token against the card details and perform a small amount of transaction (refer to this section) (which will be refunded) to validate instrument details, based on temporary token. Again, an OTP will be sent to customer and once transaction gets success, merchant will add instruments details and will get the permanent transaction token (refer to this section) which could be used later on for recurring transaction API (refer to this section) and no OTP will be sent to customer.

SCENARIO 3

Consider a situation where merchant do not want a token against card/account details and want to perform a transaction with the instrument details on the fly. In this case, a simple transaction API (refer to section API Transaction) will serve the purpose. In this process, merchant would call "validate" (refer to section API Customer Account Validation) API to get the OTP for customer and it will be sent with other transaction parameter.

API End Points

This section contains the details of all APIs provided by PayFast. These APIs could be called by the merchants, acquirers and/or aggregators. These API'S are based on REST architecture and serve standard HTTP codes for the response payload.

Get Access Token

Following API will provide you the Authentication token, which will be used to call APIs. Merchant_id and Secured_key is mandatory to get the access token. This token will be sent on all the APIs with standard HTTP header 'Authorization'.

```
curl -X POST \
  /token \
  -H 'content-type: application/x-www-form-urlencoded' \
  -d 'merchant_id=<MERCHANT_ID>&grant_type=client_credentials&secured_key=<KEY>'
```

The above command following JSON structure:

```
{
  "token": "<tokens>",
  "refresh_token": "<refresh tokens>",
  "code": "",
  "message": null,
  "expiry": <no.ofseconds>
}
```

HTTP REQUEST

POST /token

POST PARAMETERS

Parameter	Required	Description
merchant_id	Yes	Merchant ID
secured_key	Yes	Secured Key
grant_type	Yes	Grant type (client_credentials)
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

Refresh Access Token

Any access token can be refreshed upon expiry. A refresh token is given along with original token.

```
curl -X POST \
  <BASE_URL>/refreshToken \
  -H 'cache-control: no-cache' \
  -H 'content-type: application/x-www-form-urlencoded'
```

The above command returns JSON structured like this:

```
{
  "token": "<tokens>",
  "refresh_token": "<refresh token>",
  "code": "",
  "message": null,
  "expiry": <no.ofseconds>
}
```

HTTP REQUEST

POST /refreshtoken

HTTP HEADER

Authorization: Bearer <token>

POST PARAMETERS

Parameter	Required	Description
refresh_token	Yes	Refresh token
grant_type	Yes	Grant type (refresh_token)
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

Issuer/Banks

This API will provide the available list of issuer/bank.

```
curl -X GET \
  <BASE_URL>/list/banks \
  -H 'cache-control: no-cache' \
  -H 'content-type: application/x-www-form-urlencoded'
```

The above command returns following JSON structure:

```
{
  "banks": [
    {
      "bank_code": "xxxxxxx",
      "name": "xxxxxxxxxxxxx",
      "recurring_allowed": true/false,
      "otp_required": true/false
    },
    ...
  ]
  "code": null,
  "message": null
}
```

HTTP REQUEST

GET /list/banks

HTTP HEADER

Authorization: Bearer <access token>

QUERY PARAMETER

Parameter	Required	Description
reserved_1	Conditional	Reserved Field 1
customer_ip	Yes	IP address of the end user
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version

Payment (Instrument) Types

This API will provide the Payment type (or account type, e.g. Account, Wallet or Debit Card) based on selected issuer/bank.

```
curl -X GET \
  '<BASE_URL>/list/instruments?bank_code=<bank code>' \
  -H 'cache-control: no-cache' \
  -H 'content-type: application/x-www-form-urlencoded'
```

The above command returns following JSON structure:

```
{
  "bankInstruments": [
    {
      "id": "xxxx",
      "name": "xxxxxxx"
    },
  ],
  "code": "00",
  "message": null
}
```

HTTP REQUEST

GET /list/instruments

HTTP HEADER

Authorization: Bearer <access token>

QUERY PARAMETER

Parameter	Required	Description
bank_code	Yes	Bank Code
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

Issuer/Banks By Issuer Instrument ID

This API will provide the available list of issuer/bank based on instrument id.

```
curl -X GET \
  '<BASE_URL>/list/instrumentbanks?instrument_id=1' \
  -H 'cache-control: no-cache' \
  -H 'content-type: application/x-www-form-urlencoded'
```

The above command returns following JSON structure:

```
{
  "banks": [
    {
      "bank_code": "xxxxxxxx",
      "name": "xxxxxxxxxxxxx",
      "recurring_allowed": true/false,
      "otp_required": true/false
    },
  ],
  "code": null,
  "message": null
}
```

HTTP REQUEST

GET /list/instrumentbanks

HTTP HEADER

Authorization: Bearer <access token>

QUERY PARAMETER

Parameter	Required	Description
instrument_id	Yes	Instrument Id
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

Customer Account Validation

This API will be used if you choose to send OTP to registered mobile number of the customer with that respective Issuer/Bank.

```
# Example using a bank account

curl --location --request POST '<BASE_URL>/customer/validate' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'basket_id=<basket_id>' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'customer_mobile_no=<mobile number>' \
--data-urlencode 'customer_email_address=<email address>' \
--data-urlencode 'account_type_id=3' \
--data-urlencode 'bank_code=<bank code>' \
--data-urlencode 'cnic_number=<cnic number>' \
--data-urlencode 'account_number=<bank account number>' \
--data-urlencode 'order_date=<transaction date>' \
--data-urlencode 'otp_required=yes/no' \
--data-urlencode 'recurring_txn=yes/no'
```

The above command returns following JSON structure:

```
{
  "code": "",
  "message": "",
  "data_3ds_accurl": "",
  "data_3ds_pareq": "",
  "data_3ds_html": "",
  "data_3ds_secured": "",
  "data_3ds_gatewayrecommendation": "",
  "transaction_id": ""
}
```

HTTP REQUEST

POST /customer/validate

HTTP HEADER

Authorization: Bearer <access token>

POST PARAMETER

COMMON PARAMETERS

Parameter	Required	Description
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
customer_mobile_no	Yes	Customer's mobile number (format: 92-345XXXXXX)
customer_email_address	Yes	Customer's email address
account_type_id	Yes	Account type id / Payment instrument id
merCatCode	Yes	Merchant category code
subMerId	No	Sub merchant id (for Acquirer)
customer_ip	Yes	IP address of the end user
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
secured_hash	Conditional	Secured hash value

Parameter	Required	Description
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

CARD PAYMENT PARAMETERS: (UPI,MASTER,VISA AND PAYPAK CARDS)

Parameter	Required	Description
card_number	Yes	Card Number (without spaces/hyphens)
expiry_month	Yes	Card Expiry Month (e.g. 09 for September)
expiry_year	Yes	Card Expiry Year (e.g. 2021)
cvv	Yes	CVV Value
data_3ds_pagemode	For 3DS	Pagemode for 3DS auth. Required value SIMPLE
data_3ds_callback_url	For 3DS	Merchant's call back URL for 3DS auth.

(Refer to 3DS for 3DS verification process)

BANK ACCOUNT PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
account_title	No	Account Title
cnic_number	Yes	Customer's CNIC number registered with Bank

WALLET PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
account_title	No	Account Title
cnic_number	Yes	Customer's CNIC number registered with Bank

Initiate Transaction

This API will initiate payment/transaction request without token.

```
# Example using bank account
curl --location --request POST '<BASE_URL>/transaction' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'basket_id=<basket_id>' \
--data-urlencode 'txnamt=<total a amount>' \
--data-urlencode 'customer_email_address=<email>' \
--data-urlencode 'account_type_id=3' \
--data-urlencode 'customer_mobile_no=<mobiles>' \
--data-urlencode 'account_number=<account number>' \
--data-urlencode 'cnic_number=<cnic number>' \
--data-urlencode 'account_title=<account title>' \
--data-urlencode 'bank_code=<bank code>' \
--data-urlencode 'order_date=<transaction date>' \
--data-urlencode 'otp_required=yes/no' \
--data-urlencode 'recurring_txn=yes/no' \
--data-urlencode 'otp=<OTP>' \
--data-urlencode 'transaction_id=<transaction_id>'
```

HTTP REQUEST

POST /transaction

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:


```
{
  "status_code": "",
  "status_msg": "",
  "rdv_message_key": "",
  "basket_id": "",
  "transaction_id": "<transaction id>",
  "code": ""
}
```

COMMON PARAMETERS

Parameter	Required	Description
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
customer_mobile_no	Yes	Customer's mobile number (format: 92-345XXXXXX)
customer_email_address	Yes	Customer's email address
account_type_id	Yes	Account type id / Payment instrument id
merCatCode	Yes	Merchant category code
customer_ip	Yes	IP address of the end user
customer_ip	Yes	IP address of the end user
subMerId	No	Sub merchant id (for Acquirer)
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
secured_hash	Conditional	Secured hash value
otp	Conditional	Not required when issuer managed OTP
transaction_id	Yes	received from customer validation response
eci	Yes	Received from customer validation response
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

CARD PAYMENT PARAMETERS: (UPI,MASTER,VISA AND PAYPAK CARDS)

Parameter	Required	Description
card_number	Yes	Card Number (without spaces/hyphens)
expiry_month	Yes	Card Expiry Month (e.g. 09 for September)
expiry_year	Yes	Card Expiry Year (e.g. 2021)
cvv	Yes	CVV Value
data_3ds_pagemode	For 3DS	Pagemode for 3DS auth. Required value SIMPLE
data_3ds_callback_url	For 3DS	Merchant's call back URL for 3DS auth.

(Refer to 3DS for 3DS verification process)

BANK ACCOUNT PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

WALLET PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

Temporary Transaction Token

Get a temporary payment request token, against card, wallet or account details. This token can be used to send a tokenized transaction.

```
curl --location --request POST '<BASE_URL>/transaction/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'merchant_user_id=<merchant's user id>' \
--data-urlencode 'user_mobile_number=<user mobile number>' \
--data-urlencode 'basket_id=<basket_id>' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'account_type=<id>' \
--data-urlencode 'bank_code=<bank code>' \
--data-urlencode 'cnic_number=<cnic_number>' \
--data-urlencode 'account_number=<account_number>' \
--data-urlencode 'account_title=<account title>'
```

HTTP REQUEST

POST /transaction/token

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:

```
{
  "status_code": "*****",
  "status_msg": "*****",
  "instrument_alias": "*****",
  "instrument_token": "*****",
  "transaction_id": "<transaction id>",
  "otp_required": true/false,
  "eci": true/false,
  "data_3ds_acsurl": <3DSVerificationURL>,
  "data_3ds_pareq": <3DSRequestValue>,
  "data_3ds_html": <3DSRedirectionHTMLPage>,
  "data_3ds_secureid": <3DSecureID>,
  "data_3ds_gatewayrecommendation": <3DSRecommendation>
}
```

COMMON PARAMETERS

Parameter	Required	Description
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
instrument_alias	No	Instrument alias
account_type	Yes	Account type id/instrument type id
secured_hash	Conditional	Secured hash value
customer_ip	Yes	IP address of the end user
merCatCode	Yes	Merchant category code
subMerId	No	Sub merchant id (for Acquirer)
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2

Parameter	Required	Description
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version (optional)

CARD PAYMENT PARAMETERS: (UPI,MASTER,VISA AND PAYPAK CARDS)

Parameter	Required	Description
card_number	Yes	Card Number (without spaces/hyphens)
expiry_month	Yes	Card Expiry Month (e.g. 09 for September)
expiry_year	Yes	Card Expiry Year (e.g. 2021)
cvv	Yes	CVV Value
data_3ds_pagemode	For 3DS	Pagemode for 3DS auth. Required value SIMPLE
data_3ds_callback_url	For 3DS	Merchant's call back URL for 3DS auth.

BANK ACCOUNT PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

WALLET PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

Tokenized Transaction

Initiate a payment request based on "temporary" transaction token.

```
curl --location --request POST 'eBASE_URL>/transaction/tokenized' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'instrument_token=<instrument token>' \
--data-urlencode 'transaction_id=<transaction_id>' \
--data-urlencode 'merchant_user_id=<merchant's user id>' \
--data-urlencode 'user_mobile_number=<user mobile number>' \
--data-urlencode 'basket_id=<basket_id>' \
--data-urlencode 'order_date=<transaction date>' \
--data-urlencode 'txndesc=<description>' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'otp=<OTP>'
```

HTTP REQUEST

POST /transaction/tokenized

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:

```
{
  "status_code": "",
  "status_msg": "",
  "rdv_message_key": "",
  "basket_id": "",
  "transaction_id": "<transaction id>",
  "code": ""
}
```

Parameter	Required	Description
instrument_token	Yes	Instrument token
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
txndesc	Yes	Transaction description
otp	Conditional	Not required when issuer managed OTP
transaction_id	Yes	Received from get transaction token API
eci	Yes	Received from customer validation response
customer_ip	Yes	IP address of the end user
merCatCode	Yes	Merchant category code
subMerId	No	Sub merchant id (for Acquirer)
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
data_3ds_secureid	3DS	required-3DS Only
data_3ds_pares	3DS	required-3DS Only
secured_hash	Conditional	Secured hash

Add Permanent Payment Instrument

This API will allow merchants to add permanent payment instrument on PayFast. A token value will be returned to initiate the transaction for this instrument. (Currently this API is not available due to some bank/scheme level constrains).

```
curl --location --request POST 'xBASE_URL/user/instruments' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'merchant_user_id=<merchant's user id>' \
--data-urlencode 'user_mobile_number=<user mobile number>' \
--data-urlencode 'basket_id=<basket_id>' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'account_type=<t>' \
--data-urlencode 'bank_code=<bank code>' \
--data-urlencode 'cnic_number=<cnic_number>' \
--data-urlencode 'account_number=<account_number>' \
--data-urlencode 'account_title=<account title>'
```

HTTP REQUEST

POST /user/instruments

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:

```
{
  "status_code": "*****",
  "status_msg": "*****",
  "instrument_alias": "*****",
  "instrument_token": "*****"
}
```

Parameter	Required	Description
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
instrument_alias	No	Instrument alias
account_type	Yes	Account type id/Payment type id
transaction_id	Yes	Last successfull transaction id
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
merCatCode	Yes	Merchant category code
subMerId	No	Sub merchant id (for Acquirer)
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
data_3ds_secureid	3DS	required-3DS Only
secured_hash	Conditional	Secured hash value

CARD PAYMENT PARAMETERS: (UPI,MASTER,VISA AND PAYPAK CARDS)

Parameter	Required	Description
card_number	Yes	Card Number (without spaces/hyphens)
expiry_month	Yes	Card Expiry Month (e.g. 09 for September)
expiry_year	Yes	Card Expiry Year (e.g. 2021)
cvv	Yes	CVV Value

BANK ACCOUNT PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

WALLET PAYMENT PARAMETERS

Parameter	Required	Description
bank_code	Yes	Bank code
account_number	Yes	Bank account number
cnic_number	Yes	Customer's CNIC number registered with Bank
account_title	No	Account's title

OTP for Recurring Transaction

This API will generate an OTP and send to customer, which could be used for sending recurring transaction. OTP will be sent by the bank to customer's mobile number registered with the bank account.

```
curl --location --request POST 'cBASE_URLs/transaction/recurring/otp' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'instrument_token=<instrument token>' \
--data-urlencode 'merchant_user_id=<merchant's user id>' \
--data-urlencode 'user_mobile_number=<mobile>' \
--data-urlencode 'basket_id=<basket_id>' \
```

```
--data-urlencode 'txnamt=<Total amount>' \
--data-urlencode 'order_date=<transaction date>'

{
  "status_code": "",
  "status_msg": "",
  "rdv_message_key": "",
  "basket_id": "",
  "transaction_id": "<transaction id>",
  "code": "",
  "eci": true/false,
  "data_3ds_acsurl": <3DSVerificationURL>,
  "data_3ds_pareq": <3DSRequestValue>,
  "data_3ds_html": <3DSRedirectionHTMLPage>,
  "data_3ds_secureid": <3DSecureID>,
  "data_3ds_gatewayrecommendation": <3DSRecommendation>
}
```

HTTP REQUEST

POST /transaction/recurring/otp

HTTP HEADER

Authorization: Bearer <access token>

Parameter	Required	Description
instrument_token	Yes	Instrument token
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
customer_ip	Yes	IP address of the end user
cvv	Yes	CVV (for "visa/master card" based transaction)
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
secured_hash	Conditional	Secured hash

Initiate Recurring Transaction

Initiate payment transaction based on permanent token. This API will serve recurring payment requests. (Currently this API is not available due to some bank/scheme level constrains).

```
curl --location --request POST 'x-base-url/transaction/recurring' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'instrument_token=<instrument tokens>' \
--data-urlencode 'transaction_id=<transaction id>' \
--data-urlencode 'merchant_user_id=<merchant's user id>' \
--data-urlencode 'user_mobile_number=<user mobile number>' \
--data-urlencode 'basket_id=<basket id>' \
--data-urlencode 'order_date=<transaction date>' \
--data-urlencode 'txndesc=<description>' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'otp=<OTP>'
```

HTTP REQUEST

POST /transaction/recurring

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:

```
{
  "status_code": "",
  "status_msg": "",
  "rdv_message_key": "",
  "basket_id": "",
}
```

```
"transaction_id": "<transaction id>",
"code": ""
}
```

Parameter	Required	Description
instrument_token	Yes	Instrument token
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
basket_id	Yes	Merchant's Basket ID/Order ID/Cart ID/Reference ID
cvv	Yes	cvv (for "visa/master card" based transaction) (string)
txnamt	Yes	Transaction total amount
order_date	Yes	Merchant's order date (format: YYYY-MM-DD HH:mm:ss)
txndesc	Yes	Transaction description
otp	Conditional	Not required when issuer managed OTP
transaction_id	Yes	Received from get transaction token API
eci	Yes	Received from customer validation response
customer_ip	Yes	IP address of the end user
merCatCode	Yes	Merchant category code
subMerId	No	Sub merchant id (for Acquirer)
subMerName	No	Sub merchant name (for Acquirer)
subMerAbbr	No	Sub merchant abbreviation (for Acquirer)
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
data_3ds_secureid	3DS	required-3DS Only
data_3ds_pares	3DS	required-3DS Only
secured_hash	Conditional	Secured hash

Fetch Permanent Instrument Token

This API will fetch a list of stored permanent instrument tokens against provided merchant's users information. (Currently this API is not available due to some bank/scheme level constrains).

```
var client = new RestClient("<BASE_URL>/user/instruments?merchant_user_id=<merchant's user id>
&user_mobile_number=<user mobile number>");
client.Timeout = -1;
var request = new RestRequest(Method.GET);
request.AddHeader("Content-Type", "application/x-www-form-urlencoded");
request.AddHeader("Authorization", "Bearer <access token>");
IRestResponse response = client.Execute(request);
Console.WriteLine(response.Content);
```

HTTP REQUEST

GET /user/instruments

HTTP HEADER

Authorization: Bearer <access token>

```
{
  {
    "instrument_token": "*****",
    "account_type": "",
    "description": "",
    "instrument_alias": "*****"
  }
}
```

Parameter	Required	Description
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
secured_hash	Conditional	Secured hash

Delete Instrument

This API will allow merchants to delete stored permanent instrument tokens.

```
curl --location --request DELETE '<BASE_URL>/user/instruments' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>' \
--data-urlencode 'merchant_user_id=<merchant user id>' \
--data-urlencode 'user_mobile_number=<user mobile number>' \
--data-urlencode 'instrument_alias=<instrument alias>' \
--data-urlencode 'instrument_token=<instrument token>'
```

The above command returns following JSON structure:

```
{
  "code": Error/SuccessCode,
  "message": ResponseMessage,
}
```

HTTP REQUEST

DELETE /user/instruments

HTTP HEADER

Authorization: Bearer <access token>

Parameter	Required	Description
merchant_user_id	Yes	Merchant's customer/user id
user_mobile_number	Yes	Merchant's customer/user id
instrument_token	Yes	Instrument token
instrument_alias	No	Instrument alias
customer_ip	Yes	IP address of the end user
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version
secured_hash	Conditional	Secured hash

Refund Transaction Request

This API will allow merchant to initiate the request for transaction refund in case of any dispute in the transaction.

```
curl --location --request POST '<BASE_URL>/transaction/refund/<transaction_id>' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer a68adea8cc508db888625bd9a4b9d850521219df68cbf33926c4bbff8c50c00f' \
--data-urlencode 'txnamt=<total amount>' \
--data-urlencode 'refund_reason=Refund'
```


HTTP REQUEST

POST /transaction/refund/<transaction_id>

HTTP HEADER

Authorization: Bearer <access token>

The above command returns following JSON structure:

```
{
  "code": Error/SuccessCode,
  "message": ResponseMessage,
}
```

Parameter	Required	Description
transaction_id	Yes	Successfull transaction id to be refunded
txnamt	Yes	Original transaction amount
customer_ip	Yes	IP address of the end user
refund_reason	Yes	Reason of refund
reserved_1	Conditional	Reserved Field 1
reserved_2	Conditional	Reserved Field 2
reserved_3	Conditional	Reserved Field 3
api_version	No	API Version

Get Transaction Status

This API will provide transaction status with respect to the transaction id or the merchant's provided basket id/order id.

```
// Example code for getting transaction status
// using transaction id

curl --location --request GET '<BASE_URL>/transaction/<transaction_id>' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Bearer <access token>'
```

The above command returns following JSON structure:

```
{
  "status_code": "",
  "status_msg": "",
  "rdv_message_key": "",
  "basket_id": "",
  "transaction_id": "<transaction id>",
  "code": ""
}
```

HTTP REQUEST

TRANSACTION STATUS BY TRANSACTION ID

GET /transaction/<transaction_id>

HTTP HEADER

Authorization: Bearer <access token>

Parameter	Required	Description
api_version	No	API Version

TRANSACTION STATUS BY BASKET ID/ORDER ID (LAST)

GET /transaction/basket_id/<basket_id>

HTTP HEADER

Authorization: Bearer <access token>

Parameter	Required	Description
order_date	Yes	Transaction/Order Date
customer_ip	Yes	IP address of the end user

Parameter	Required	Description
api_version	No	API Version

HTTP HEADER

Authorization: Bearer <access token>

Error Codes

CODES		DESCRIPTION
00	Processed OK	
002	Time Out	
001	Pending	
97	Dear Customer, you have an insufficient Balance to proceed	
106	Dear Customer, Your transaction Limit has been exceeded please contact your bank	
3	You have entered an Inactive Account	
14	Entered details are Incorrect	
55	You have entered an Invalid OTP/PIN	
54	Card Expired	
13	You have entered an Invalid Amount	
126	Dear Customer your provided Account details are Invalid	
75	Maximum PIN Retries has been Exceeded	
14	Dear Customer, You have entered an In-Active Card number	
15	Dear Customer, You have entered an In-Active Card number	
42	Dear Customer, You have entered an Invalid CNIC	
423	Dear Customer, We are unable to process your request at the moment please try again later	
41	Dear Customer, entered details are Mismatched	
801	{0} is your PayFast OTP (One Time Password). Please do not share with anyone.	
802	OTP could not be sent. Please try again later.	
803	OTP has been sent to your email address.	
804	OTP has been sent to your mobile number.	
805	OTP Verified	
806	OTP could not be verified.	
807	Too many attempts. Please try again later in few minutes.	
808	Passwords do not match	
809	Invalid Password	
810	Password could not be changed	
811	Password changed successfully	
812	Request could not be validated. Please try again.	
813	Email address already registered	
850	OTP not required because issuer manages OTP itself.	
851	OTP required for permanent token.	
79	Alternate Success response	
9000	Rejected by FRMS	

Hashed Parameters

API Reference	Parameters to be Hashed	Data
Temp. Token For Transaction	merchant_user_id	For Card :- merchant_user_id+ user_mobile_number+card_number+ expiry_month+expiry_year+cvv
	user_mobile_number	For Account :- merchant_user_id+ user_mobile_number+account_number+ cnic_number
Card	card_number	
	expiry_month	For Wallet: - merchant_user_id+ user_mobile_number+account_number+ cnic_number
	expiry_year	
	cvv	
Account	account_number	
	cnic_number	
Wallet	account_number	
	cnic_number	
Temp. Token Transaction	instrument_token	instrument_token + merchant_user_id + user_mobile_number + txnamt + otp
	merchant_user_id	
	user_mobile_number	
	txnamt	
	otp	
OTP for Recurring Transaction	instrument_token	instrument_token + merchant_user_id + user_mobile_number
	merchant_user_id	
	user_mobile_number	
Initiate Recurring Transaction	instrument_token	instrument_token + merchant_user_id + user_mobile_number + txnamt + basket_id
	merchant_user_id	[+ otp]
	user_mobile_number	
	txnamt	
	basket_id	
	otp (if applicable)	
Customer Account Validation		
Common	basket_id	
	txnamt	
Card	card_number	basket_id + txnamt + card_number+ expiry_month+expiry_year+cvv
	expiry_month	
	expiry_year	basket_id + txnamt + account_number + cnic_number
	cvv	
Account	account_number	basket_id + txnamt + account_number + cnic_number
	cnic_number	
Wallet	account_number	
	cnic_number	
Initiate Transaction without Token		
Common	basket_id	

API Reference	Parameters to be Hashed	Data
	txnamt	
	otp	
Card	card_number	basket_id + txnamt + card_number+ expiry_month+expiry_year+cvv+otp
	expiry_month	
	expiry_year	basket_id + txnamt + account_number + cnic_number+otp
	cvv	
Account	account_number	basket_id + txnamt + account_number + cnic_number+otp
	cnic_number	
Wallet	account_number	
	cnic_number	
	txnamt	
	otp	
Add Permanent Instrument		
	merchant_user_id	merchant_user_id+ user_mobile_number+card_number+ expiry_month+expiry_year+cvv
	user_mobile_number	merchant_user_id+ user_mobile_number+account_number+ cnic_number
Card	card_number	
	expiry_month	merchant_user_id+ user_mobile_number+account_number+ cnic_number
	expiry_year	
	cvv	
Account	account_number	
	cnic_number	
Wallet	account_number	
	cnic_number	
Fetch Permanent Instruments		
	merchant_user_id	merchant_user_id+ user_mobile_number
	user_mobile_number	
Delete Stored Instrument		
	merchant_user_id	merchant_user_id + user_mobile_number + instrument_token
	user_mobile_number	
	instrument_token	

Note:

1. "+" symbol denotes simple string concatenation.
2. Hashed data will be received in "secured_hash" parameter in designated APIs
3. A key will be provided separately for the hash calculation.
 - Hash Method: HMAC (Keyed Hash)
 - Hash Algorithm: sha256
 - Hash Key: ASCII encoded string |

3D Secure Transaction

In case of Visa/Master card transaction, 3DS verification will be performed.

A. APIs Customer Account Validation, Get Temp. Transaction Token, OTP for Recurring Transaction, will accept an additional parameter `data_3ds_pagemode`, it has 1 value: `SIMPLE`.

i. In `SIMPLE` case, `data_3ds_html`, will contain a pre-formatted html page which can be used directly to redirect customer to verify OTP. You are not required to modify this value in any case. Any modification in this value may lead to unexpected results.

B. APIs Initiate Transaction without Token, Temp. Token Transaction, Initiate Recurring Transaction, will accept 2 more additional parameters for 3DS verification:

i. `data_3ds_secureid`: This will be returned in response of APIs Customer Account Validation, Get Temp. Transaction Token, OTP for Recurring Transaction (Customer Validation).

ii. `data_3ds_pares`: This value will be returned in call back URL which is called after 3DS verification, the parameter is returned `paRes`.